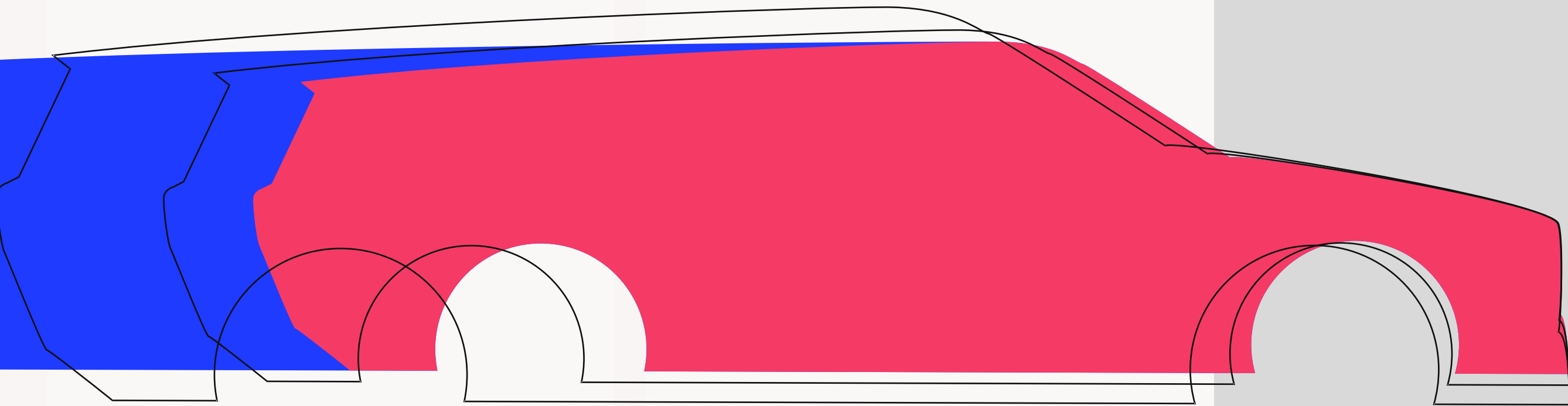


Thermofluids

Individual Submission

PAGE 1



Selected Vehicle Archetype: **Estate Car**

Vehicle Archetype

DE2 Thermofluids - Part 1

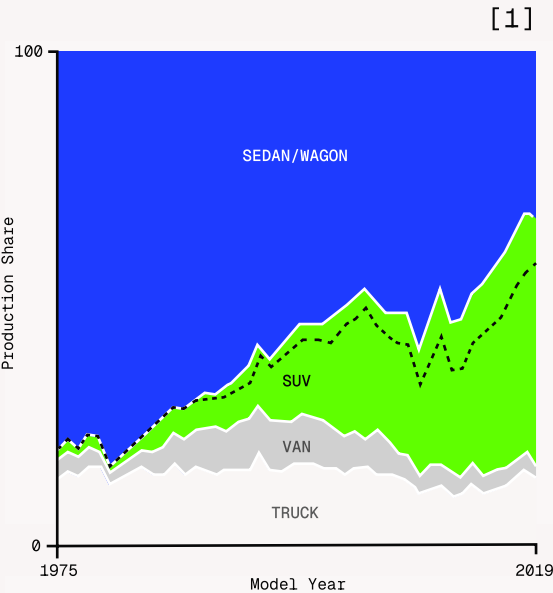
Estate Cars / Station Wagons

Estate cars are practical, personal vehicles typically based on a saloon (sedan) platform. They share the same front structure but extend rearwards to create a larger, more versatile load space behind the rear seats.

Typical Users

The increased space and practicality of estate cars makes them well suited to families needing flexibility, as well as active users transporting gear or luggage.

Key priorities for estate car users include interior space, stability across a range of speeds, and fuel/battery efficiency. In the context of electric vehicles, efficiency is increasingly important, enabling longer journeys and reducing range anxiety.



+21%
Fuel Efficiency
over SUVs

[1]

Improving Efficiency

$C_d \downarrow$

Users want the car to be efficient, whether in fuel or battery use. This can be achieved by designing the body to minimise drag, helping the vehicle move more easily through the air.



Selection Rationale

Family-focused, practical cars make up the largest share of vehicle sales. However, there is a noticeable lack of estate or wagon options in today's market (6%), particularly when compared to the growing number of crossovers and SUVs. This trend is even more apparent in the EV sector, with only 8 electric estate models being available in the UK.

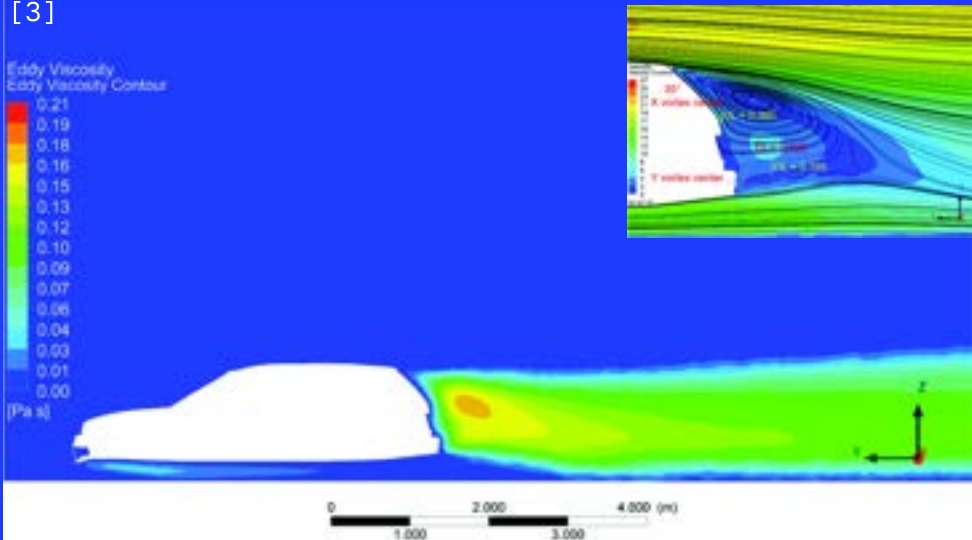
Estate cars' lower profile also offer an inherent aerodynamic advantage, while still maintaining a high level of practicality. This makes them a category worth optimising, as they have the potential to provide a more efficient and effective alternative to the market-dominant crossover.

Creating Stability

$-C_L$

People want a car that feels stable and planted. To achieve this we can give the vehicle an overall aerodynamic profile that generates a slight, consistent downforce.

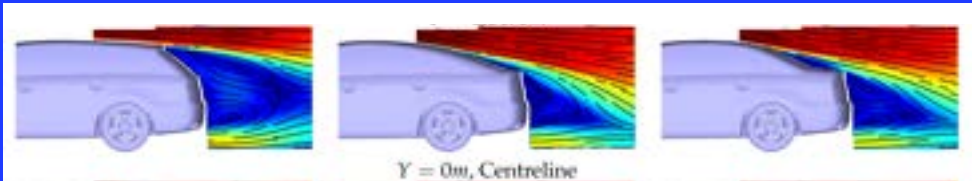
[3]



Current Issues

Previous analysis of cars in this category show that the most common problems are presented in the wake of the vehicle - the nature of the large boot/hatch mean that there will be a larger area of negative pressure at the rear of the vehicle.

For these 'Square-back' cars, the 'pull' on the back of the car caused by static low pressure far out weighs the problems with high pressure 'pushing' on the front of the car. Most aerodynamic efficiency gains will likely be found in this area.



[4]

Design Objectives

Derived from User Desires

Primary Objective: Reduce Drag
Secondary Objective: Generate Downforce

Practical Design Requirements

- Maintained interior space suitable for people and cargo
- Clear outward visibility
- Easy to manoeuvre, especially at low speeds
- Practical access with well-sized doors and openings
- Suitable ground clearance for everyday use

Benchmarking + Techniques

DE2 Thermofluids - Part 1

Aerodynamic Ideal

For ideal aerodynamic performance, the vehicle will have a fully attached flow of air across the full length of the body. Flow separation should occur at the rear of the car, distanced from the body as much as possible.

Minimal Displacement Attached Flow Flow Separation

VW ID.7 TOURER

Production
Estate Car
BEV



REAR SEPARATION EDGES

These benefit the efficiency by reducing the ‘wrap around’ effect as the flow attached to the side + top profiles of the car reaches the end of the body. The sharp extended edges force the flow of air to separate, reducing the ‘pull back’ effect of low pressure behind the vehicle.

MERCEDES EQXX

Concept
Saloon
BEV

[5]

EXTREME LOW DRAG OPTIMISATIONS

The EQXX is an extremely aero efficient vehicle, achieving a drag coefficient of 0.17. In addition to the smooth + sleek surfaces, the car uses an underfloor optimised for passive battery cooling and an active diffuser that adapts the downforce generated at high speeds.

PORSCHE TAYCAN SPORT TURISMO

Production
Estate Car
BEV

[6]

FRONT WHEEL AIR CURTAINS

To reduce the turbulent impact of the wheels, air is funnelled and pushed from the front of the vehicle around and over the face of the wheel. This creates a sealing effect, reducing the flow of air that enters/exits the wheel-arch, this allows the flow to remain ‘attached’ over the side profile.

HONDA INSIGHT (1999)

Production
Hatchback Coupe
ICE

[7]

WHEEL SPATS + PRESSURE RELEIF

The distinctive feature of this car is the rear wheel spats, which cover the wheel-arch and allow air to flow more smoothly over the rotating surface. It also features aggressive body ‘cuts’ behind the front wheels, reducing the ‘pullback’ effect of the low pressure zone that forms behind the wheel.

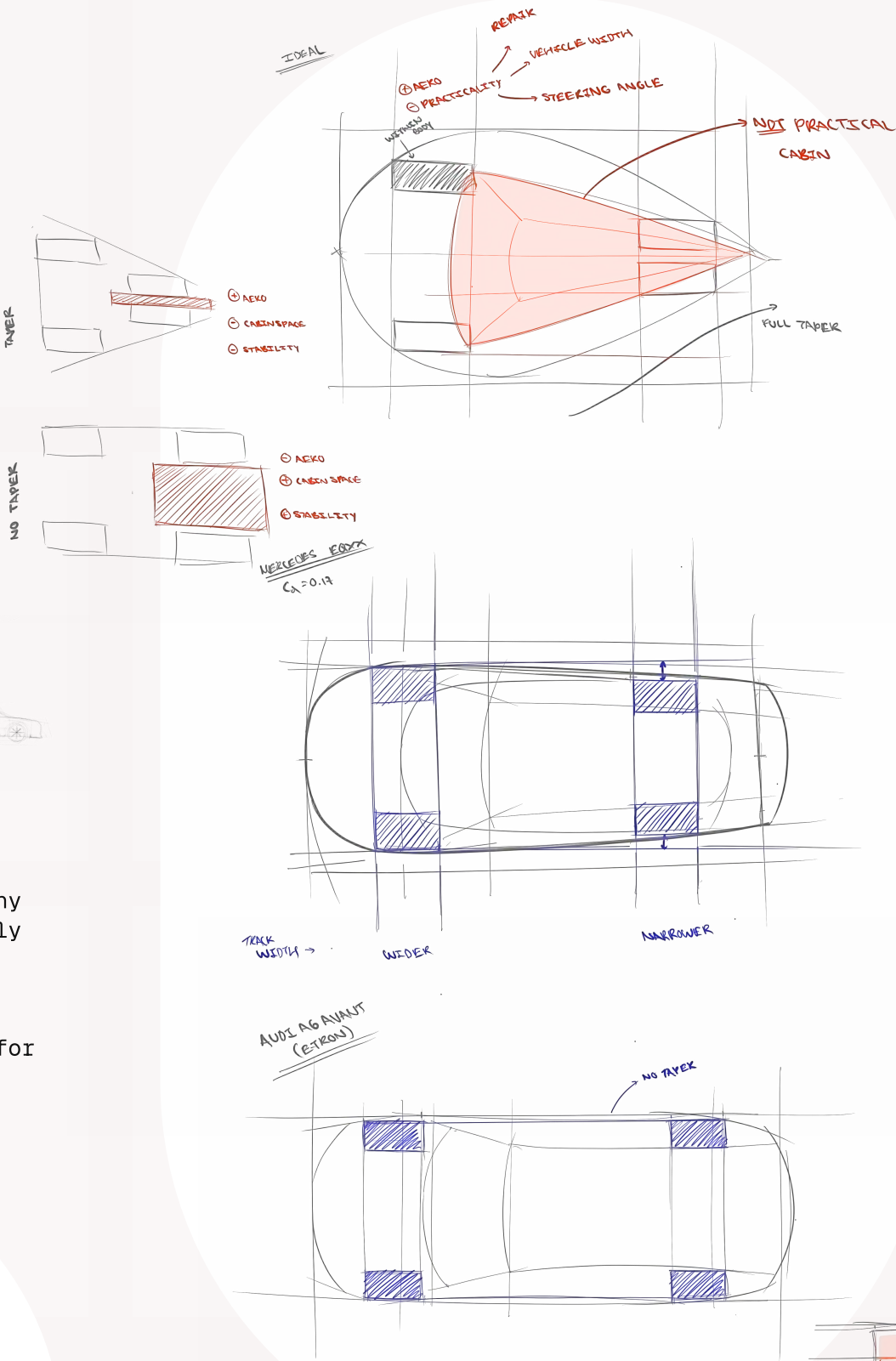
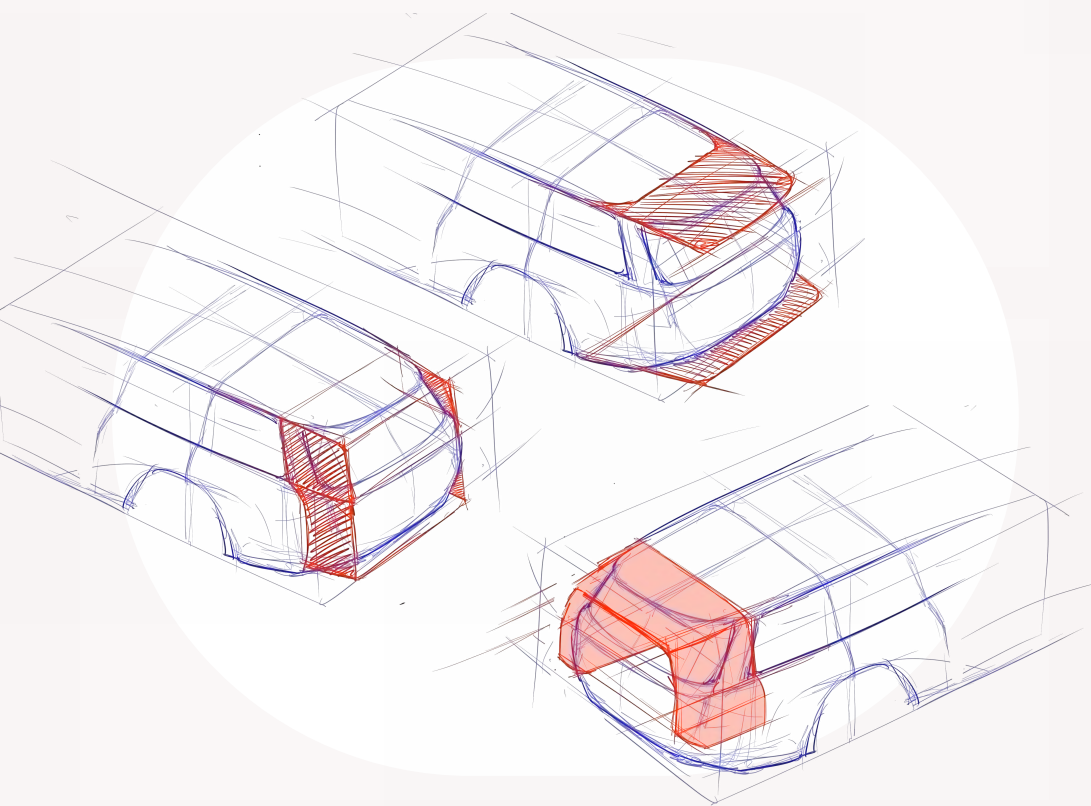
Ideation

DE2 Thermofluids - Part 1



EXTENDED REAR SEPARATION SURFACES

'Sharp' separation edges are already common on many estate cars and hatchback vehicles, by dramatically extending them, we can further continue sloping edges, reducing rear surface area and therefore wake. These edges could be extended below the window-line or cover a greater perimeter portion for more coverage.

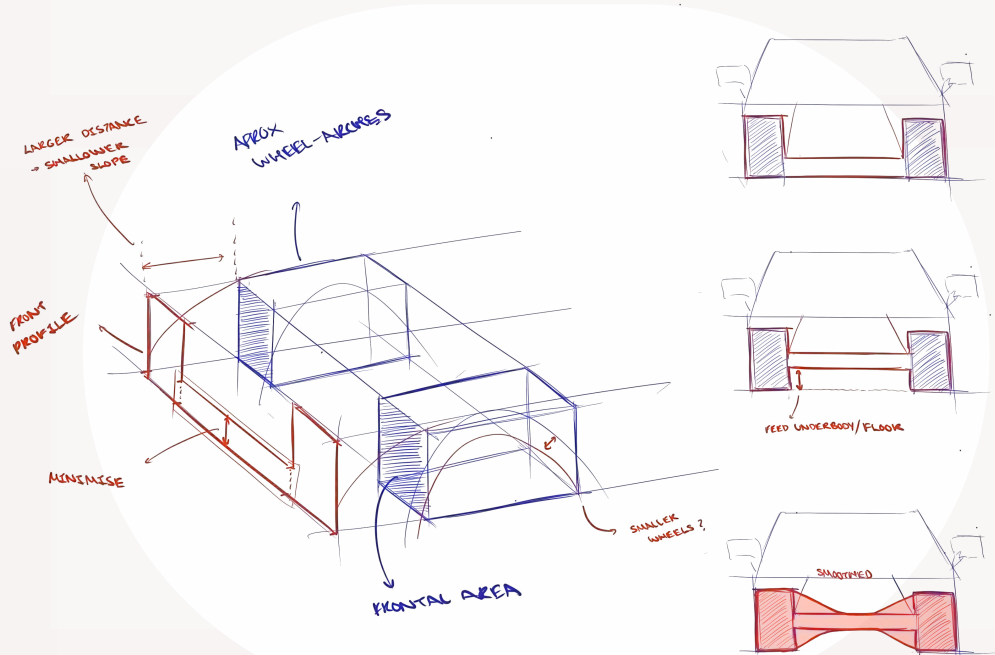


WHEEL TAPER + VEHICLE TRACK WIDTH

Arranging wheels inside the Aerodynamic Ideal (teardrop) shape, the complete taper forces the rear wheels together. The Mercedes EQXX, although less obvious, also features a taper in track-width from front to back, but keeps a wide stance for high speed stability. In contrast, the more conventional AUDI A6 AVANT E-TRON has no visible taper.

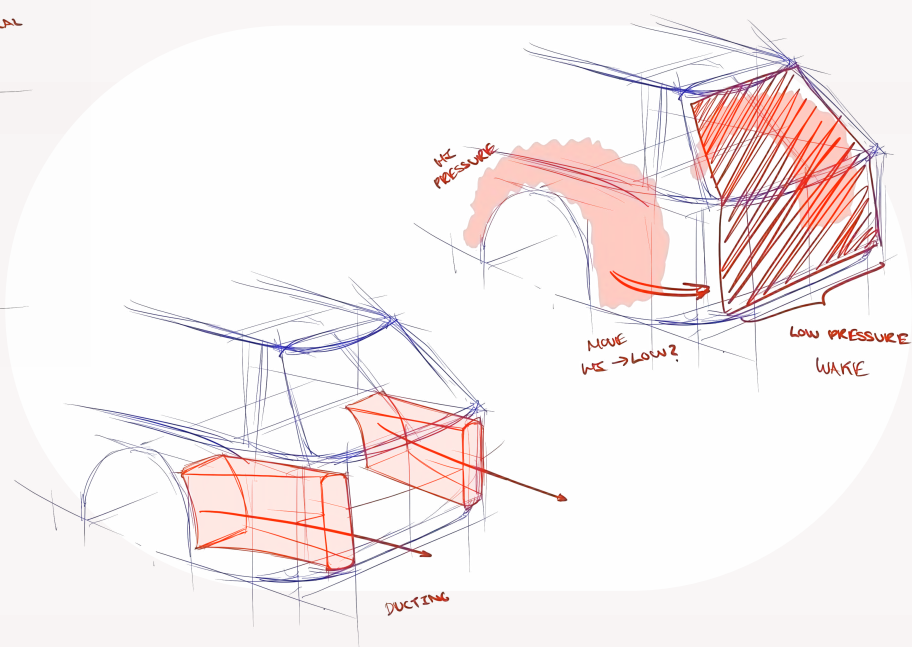
MINIMISING FRONTAL AREA

Having the smallest possible frontal area is essential to reducing the drag force. By looking at a frontal profile, I determined that the minimum frontal area would encompass the front wheels as well as a 'bar' joining them. Shifting this bar away from the road also allows more air to flow under the vehicle.



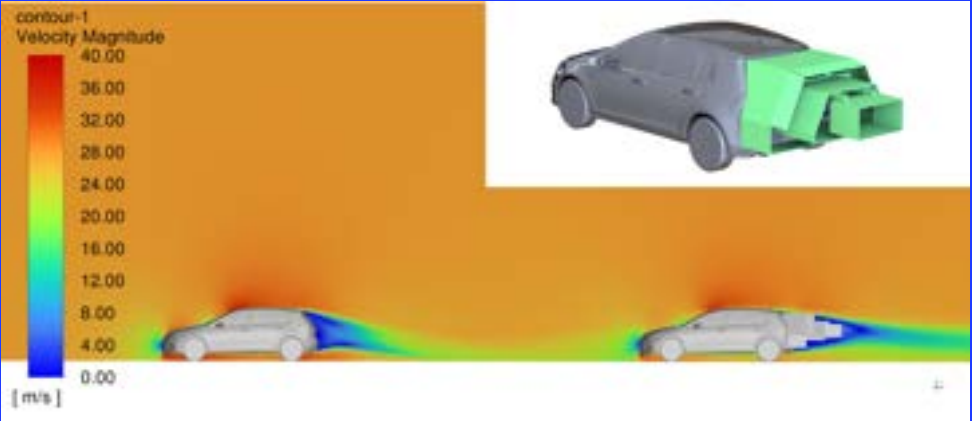
DUCTING TO REDUCE WAKE

To combat the large wake created by the estate arch, we could use the buildup of high pressure in the wheel-arches to funnel air to the low pressure zone created behind the vehicle. Ducts from the back wheels to the rear surface may be able to redistribute this pressure and reduce the wake.



Further Ideation

DE2 Thermofluids - Part 1

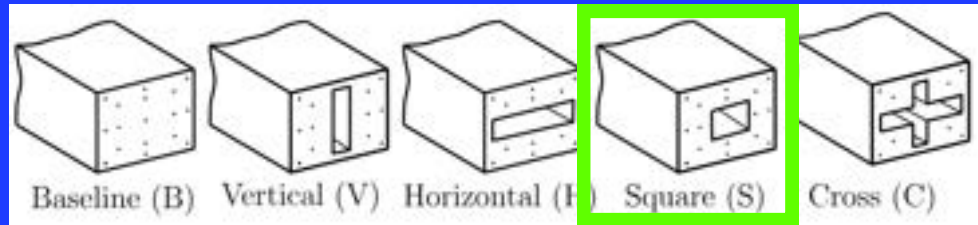


[9]

MULTI-ELEMENT SEPARATION SURFACES

Multi-element aero surfaces are common across spoilers, diffusers and splitters. However, they are not commonly used for the application of drag reduction. Despite the unpopularity, the above paper [9] shows that they can be extremely effective. This specific configuration on a VW Golf hatchback significantly shifting wake and decreasing drag coefficient from 0.305 to 0.236 in their simulations.

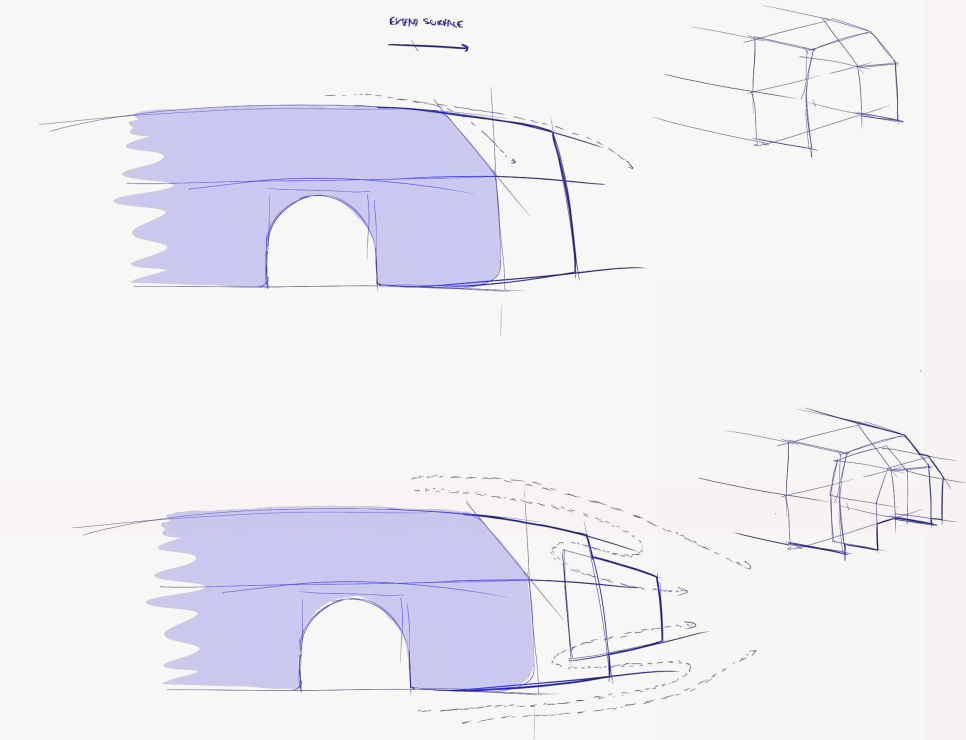
Slot			Results	
Study	Geometry	S_b/hw	$\partial\Delta C_B/\partial(C_q(hw/S_b)^{1/2})$	$\Delta C_{B,max}$
Present (S)		0.1	-0.323	-0.0238
Lorite-Díez et al. [2020]		0.035	-0.181	-0.0036



[10]

DUCTING GEOMETRY

The shape and positioning of ducts at the rear of the car has a large effect on how effective they are [10]. In my original ideation the ducts were positioned at the sides of the car, near the inlets in the rear wheel-arches. However, from my research, the most effective position of the ducts is at the centre. This configuration is around 6.6x more drag-reducing.

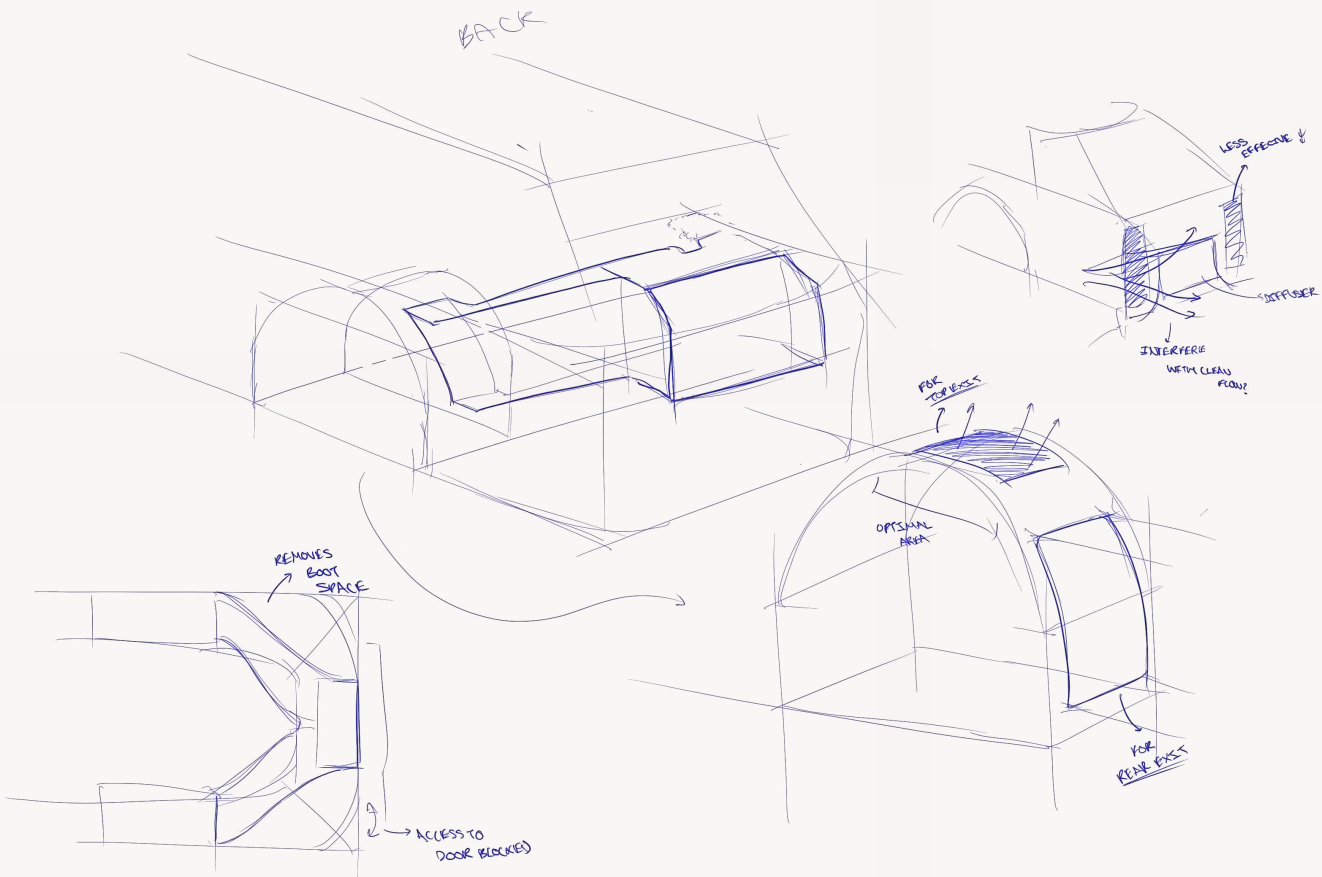
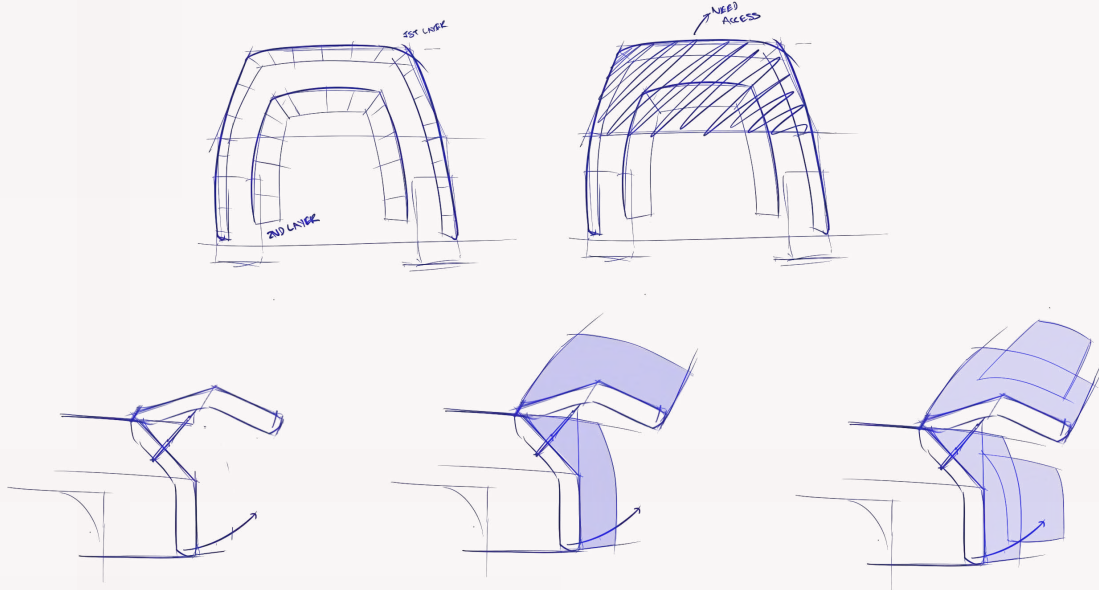


SEPARATION SURFACE GEOMETRY

The multi-element surfaces need to follow and integrate with the body lines of the vehicle. This allows the attached flow to better detach from the main vehicle body. ‘Mirroring’ this geometry with the inner surface allows the flow to separate without disruption and gives turbulent flow another surface to be guided away from the car.

CLEARANCE + REALISM

The practicality of the vehicle is also key, opting for 2 surfaces rather than 3+ meant that there would still be acceptable rear visibility. The boot opening also has to be very accessible, to address this, the surfaces should feature gaps at the bottom edge as well as limited length so that a user can get to the hatch easily.

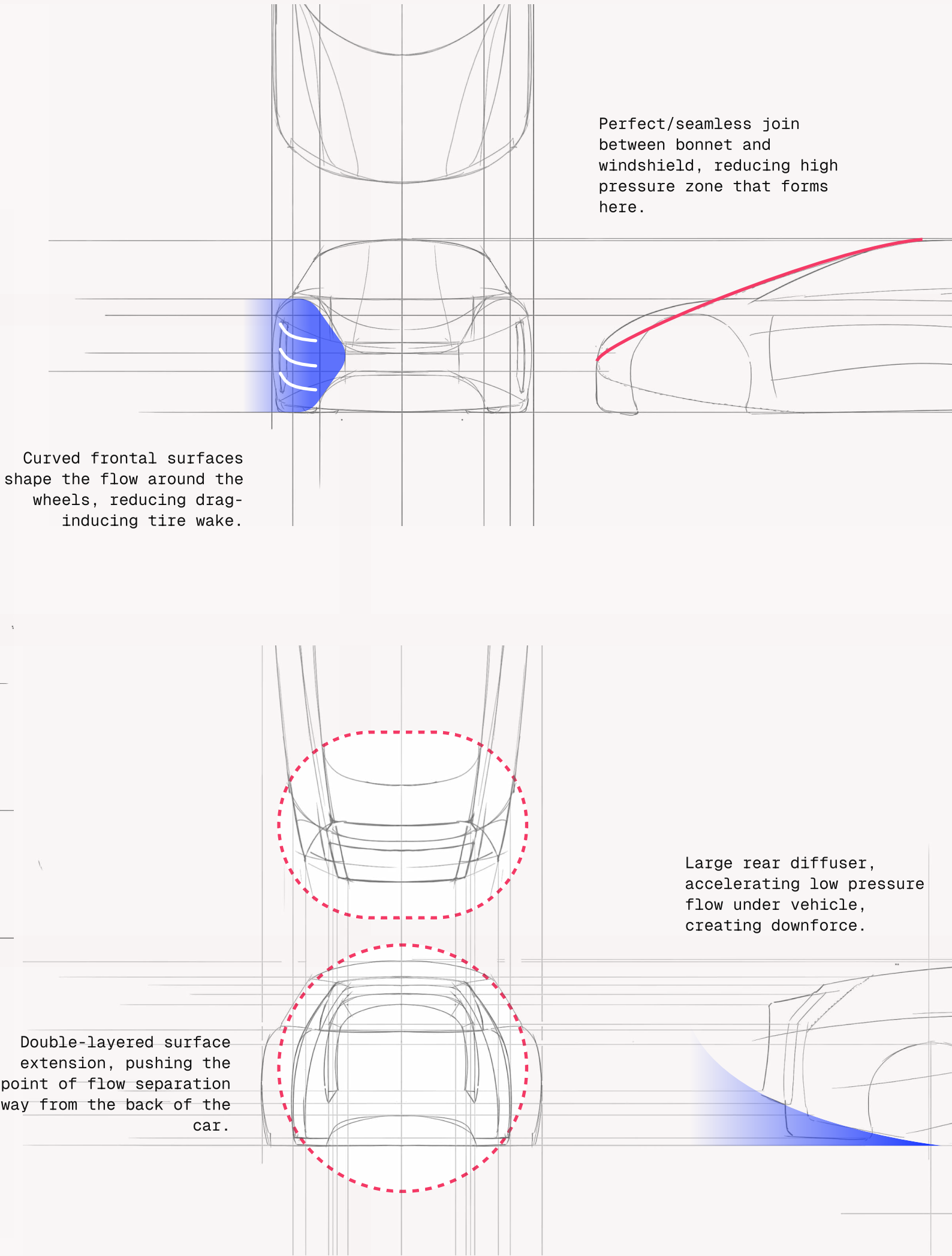
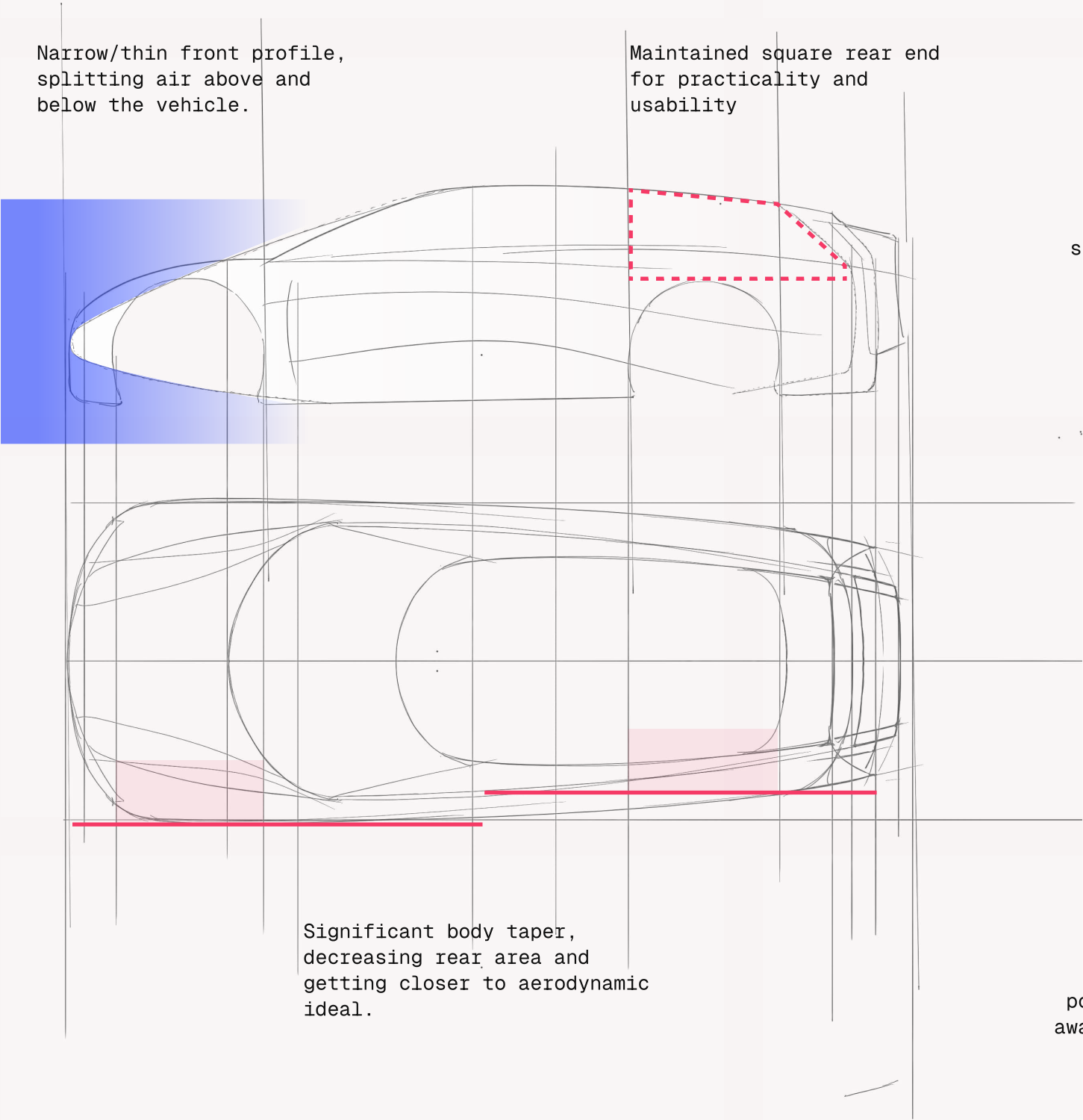


BOOT SPACE ISSUES

Moving the ducts to a central position at the back of the car is more effective, but it is also much less practical. This would basically remove all usable boot space and reduce visibility. Additionally, having the vertical ducts at the sides of the car may interfere with the diffuser, so I decided not to continue with them, focussing on the separation surfaces.

Design Summary

DE2 Thermofluids - Part 1



CAD Model

DE2 Thermofluids - Part 1

Method



Starting with the orthographic drawings of front, back, top and side profiles.



Next, a plane was extruded along the 'shoulder' line and down to the door sills.



Then, the wheel-arches were added, along with a raised section for the swept roof.



After this, the windshield was connected, extending the curve down to the front bumper.



The front and rear bumpers were then modelled, connecting the sides and top profiles.



Finishing the underbody allowed the body to become solid, then wheels were added.



The final pieces added were the double-element separation surfaces at the back.

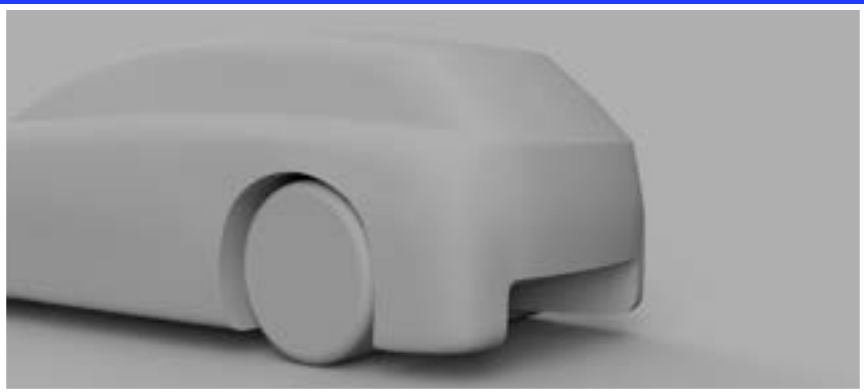
SIMPLE

Basic Body - No Added Surfaces

The basic version of of the model is an enhanced version of existing estate car geometries. The front bonnet area has a shallow angle and minimal changes in curvature.

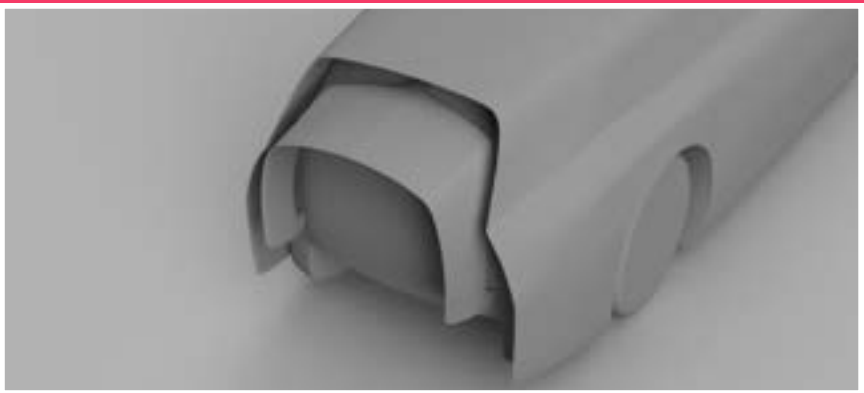
The crease of the shoulder-line tracks right from the nose to the rear in one simple curve. It rises at the front of the car, before sweeping down towards the back, acting as a down-washing element.

At the rear of the car, the diffuser exits, shielded from the wake of the rear wheels.



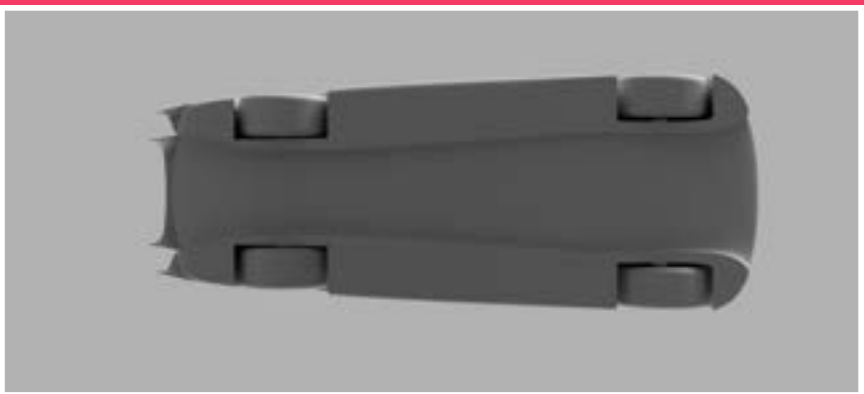
WITH SEP. SURFACES

Addition of the Rear 'Cones'



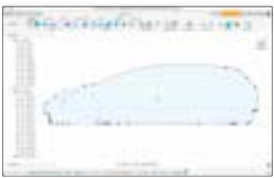
The complete version of the car model includes a double seperation surface. The first (outer) section is a direct continuation of the car's body and overall outer shape.

Inside of this, the second section follows the slimming 'taper' effect of the body, but with a simpler and more square profile. It is joined to the main body at 3 points, 2 at the bottom corners, and one in the top centre of the profile.



Dimension Calculations

	F. AREA	LENGTH	WIDTH	HEIGHT
FULL SIZE	1.8m^2	4.4m	1.7m	1.2m
SIMPLE	0.007m^2	274mm	107mm	73mm
SEP. SURF.	0.007m^2	300mm	107mm	73mm



Using my CAD model, I projected the side and frontal profiles of my design, this allowed me to find the frontal area as well as overall length, height and width. This process was repeated for full size and 1:17 scale as well as simple vs separation surface versions of the model.

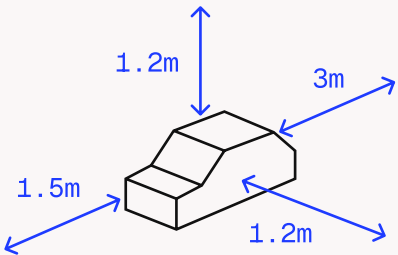
CFD Baseline

DE2 Thermofluids - Part 1

Setup

To practice and get an initial measurement of my car model, I decided to run a CFD simulation **before adding wheels and aero surfaces**. This modelling was done **at a 1:1 scale** and allowed me to identify areas on the body that were particularly bad before finishing the CAD model.

ENCLOSURE



*These values are relatively low and so may lead to a high blockage ratio, which may increase drag and lift coefficients.

MESHING

Local Sizing: 0.003 Maximum Cell Size: 0.05 Boundary Layers: 10

Total Cell Count:
9,000,000

MODEL

Viscous: k-omega SST

FLUID PROPERTIES

Fluid: air Density: 1.225 Velocity: 30.5

INITIAL + BOUNDARY CONDITIONS

car, road, walls: wall
inlet: velocity-inlet
outlet: pressure-outlet
symmetry: None

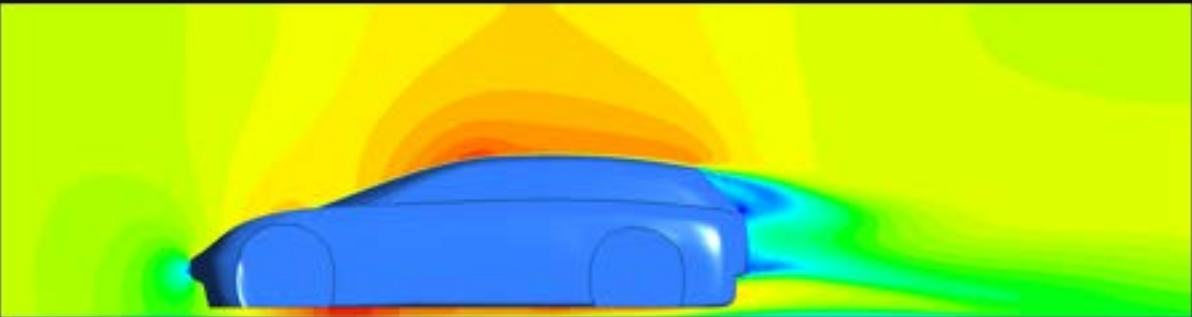
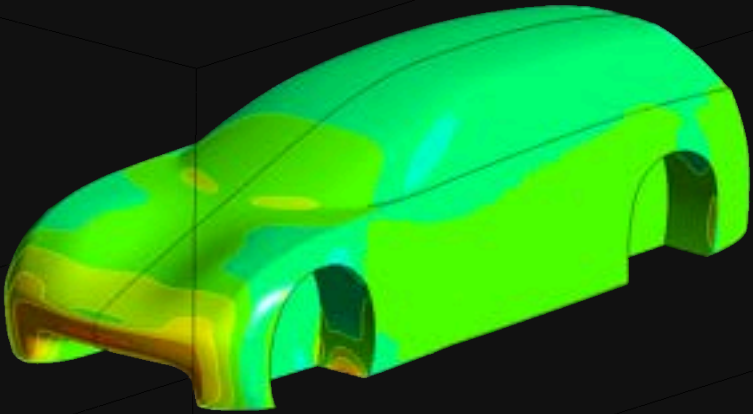
REFERENCE VALUES

Area (half): 0.9m² Length: 4.4m

Results

VEHICLE SURFACE PRESSURE

Overall, the surface pressure is good, with a minimised site of high pressure at the front of the vehicle and very few areas of extreme low pressure. The only area that caused enough problems to need modifying was the nose, where I smoothed the curvature to reduce the the concentration of high pressure.

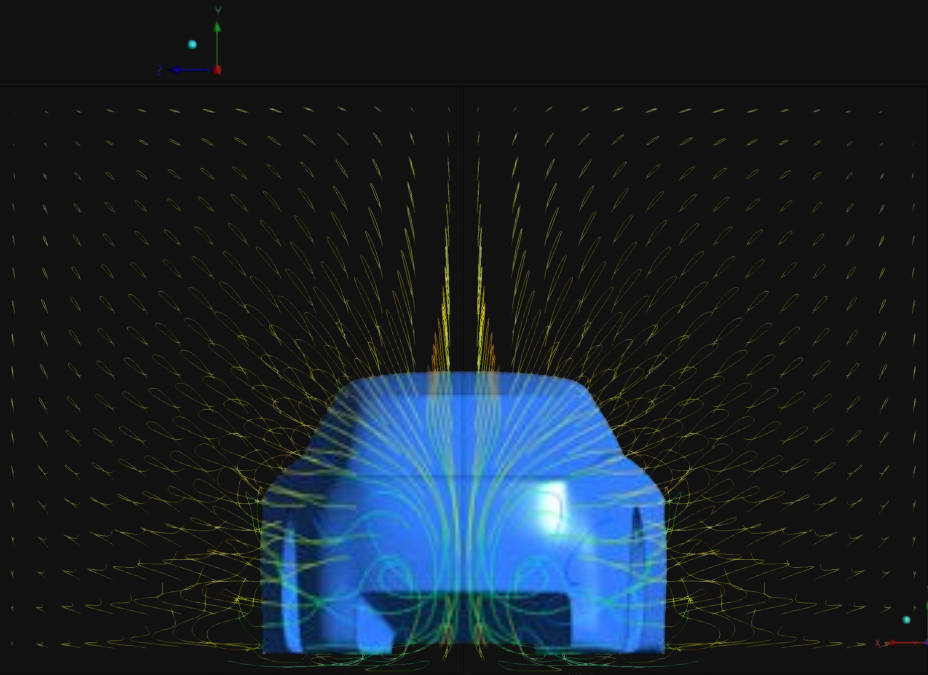


AIR VELOCITY

The velocity plot for the baseline geometry showed that many of the basic features were working as intended. The diffuser and Venturi-Effect tunnel under the car show the air speeding up before being 'released'.

WAKE - STREAMLINES

Using streamlines it is easy to see that the main point of disturbance and vorticity is the wake at the rear of the car. The flow is pulled inwards from the sides and top profile, causing spiralling and interference with the flow of air from the diffuser.



0.391



-0.11

Comparisons

DE2 Thermofluids - Part 1

Setup

The setup for these tests is almost identical to that in the baseline model. Below are the changes made:

MESHING

Local Sizing: 0.001

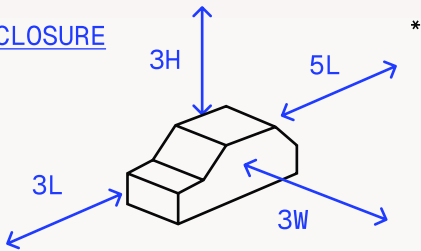
Maximum Cell Size: 0.02

Total Cell Count: 126,000

REFERENCE VALUES

Area (half): 0.0035m^2
Length: L*

ENCLOSURE



MODEL

Viscous: k-epsilon

*These values differ between the different versions of the car model so were different across the simulations.

Drag Coefficients

0.313

SIMPLE

Basic Body - No Added Surfaces.

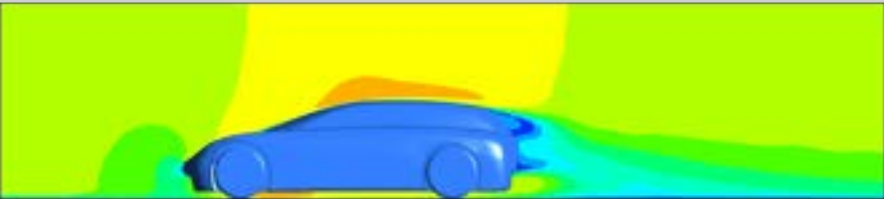
0.286

WITH SEP. SURFACES

Addition of the Rear 'Cones'.

The addition of separation surfaces has a significant impact on the overall drag coefficient, a reduction of 0.027 while only only changing the rear flow separation.

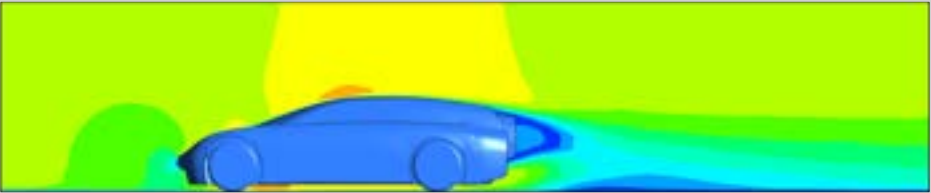
SIMPLE



AIR VELOCITY

The simple configuration matches the baseline test, you can see how the wake disrupts the flow behind the car, interfering with the flow of air out of the diffuser.

WITH SEP. SURFACES



The separation surfaces are very effective, shifting the wake upwards and away from the rear of the car, demonstrating similar effects to the research it was inspired by. []

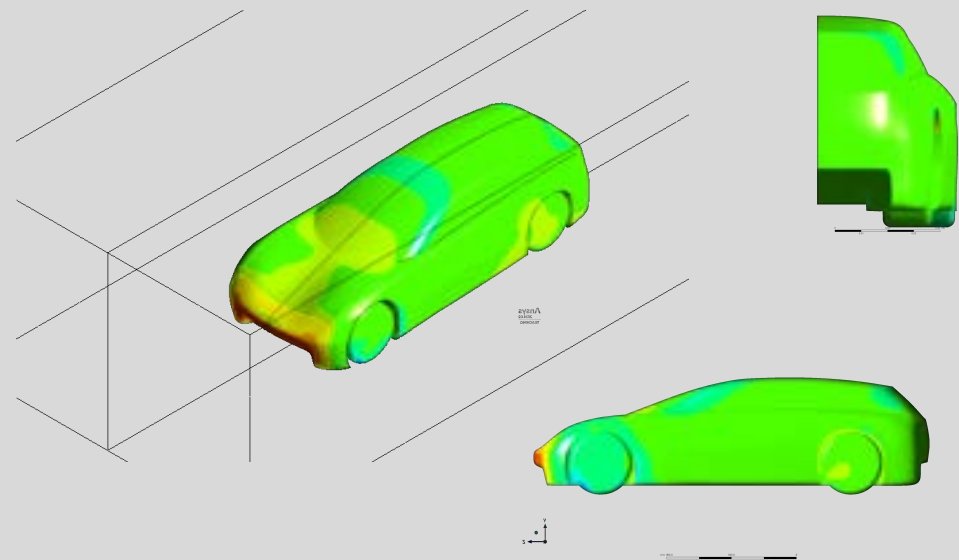
STREAMLINES

Streamline plotting clearly shows effect of the wake, the car performs well up until the rear section. The clean flow detaches violently, causing a problematic wake.

The effect of the surface extension is very visible, with the streamlines at the rear of the car having almost no vorticity or turbulence as they exit the back of the body.

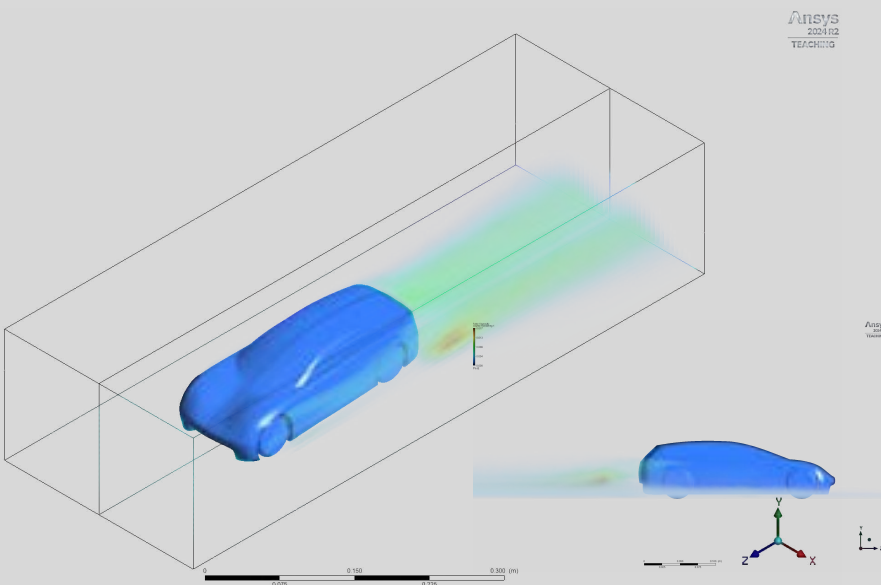
Comparisons 2

DE2 Thermofluids - Part 1



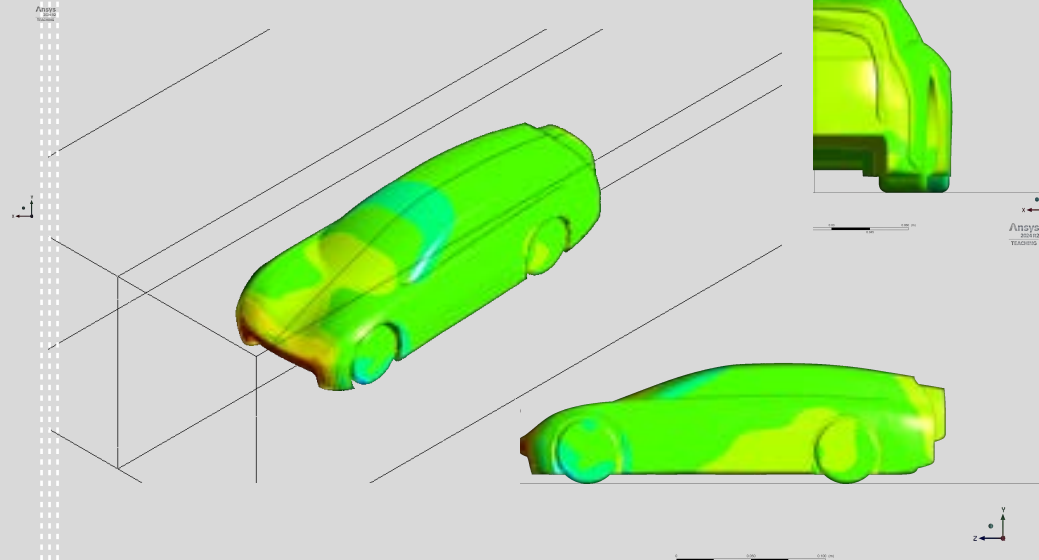
PRESSURE CONTOURS

The pressure contours show the relatively small concentrated area of high pressure (with small improvements over the 'baseline') and consistent mid-low pressure across the sides and back of the vehicle.

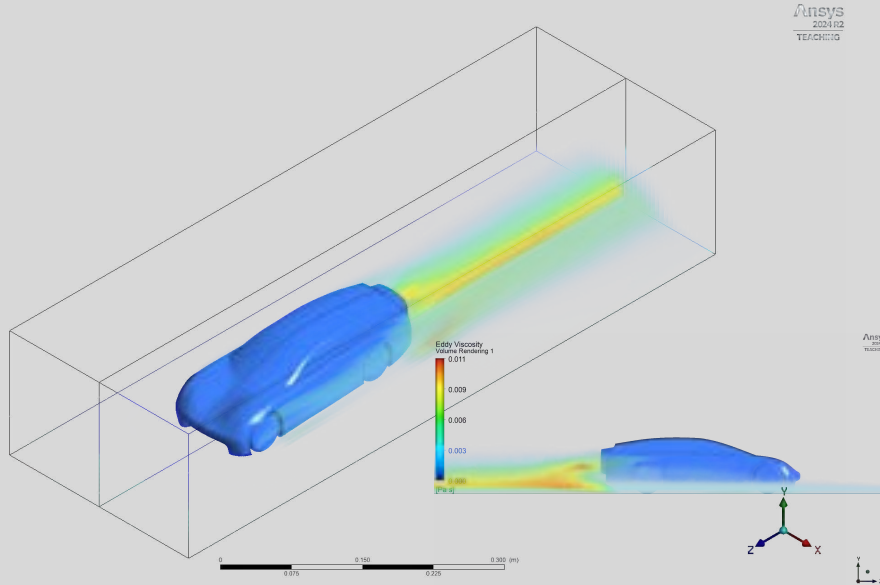


FLOW TURBULENCE - EDDY VISCOSITY

The 3D eddy plot shows that the simple version of the car has a wide area of turbulence, with a more intense section at road height separated from the vehicle, indicating the severe wake.



The separation surfaces have no effect on the pressure on the front of the vehicle. However they do increase the pressure on the body elements at the rear, which is an improvement from the overall low pressure that the simple model demonstrates.



The separation surfaces significantly reduce the area of turbulent air. The colours of the graph are a little misleading as this configuration also reduces the overall maximum eddy velocity, representing a very significant reduction in wake effects.

Lift Coefficients

SIMPLE

0.261

SEP. SURFACES

0.166

After doing my baseline testing (pg.9) and achieving a lift coefficient of -0.11, I focussed on optimising for drag. However, when adding wheel elements for the simple configuration, the lift jumps up by 0.35.

The separation surfaces are very effective at reducing the lift generated, helping to shift wake upwards and away from the diffuser. This results in a much cleaner flow of air at the rear of the car, clearly contributing to the near -0.1 in lift coefficient.

Force Calculations

Using the ANSYS calculated coefficients and the equations for drag and lift, I hand calculated the forces for both my 1:17 simulations.

$$F_D = \frac{1}{2} \rho v^2 C_D A$$

1.2483838719

1.1406958062

$$L = \frac{1}{2} \rho v^2 A C_L$$

1.04098463438

0.6620821812

Concept Development

DE2 Thermofluids - Part 1

Design Requirements

$C_d \downarrow$

Primary Objective:
Reduce Drag

$-C_L$

Secondary Objective:
Generate Downforce

Practical Design Requirements

- Maintained interior space suitable for people and cargo
- Clear outward visibility
- Easy to manoeuvre, especially at low speeds
- Practical access with well-sized doors and openings
- Suitable ground clearance for everyday use

Verdict

PRIMARY + SECONDARY OBJECTIVES

Optimising for drag throughout my iterations, and reducing it by over 0.03 from my initial designs makes me confident that I have satisfied this requirement. However, I did not manage to get a negative lift coefficient. This is something that will need to address in further improvements/redesigns.

PRACTICAL DESIGN REQUIREMENTS

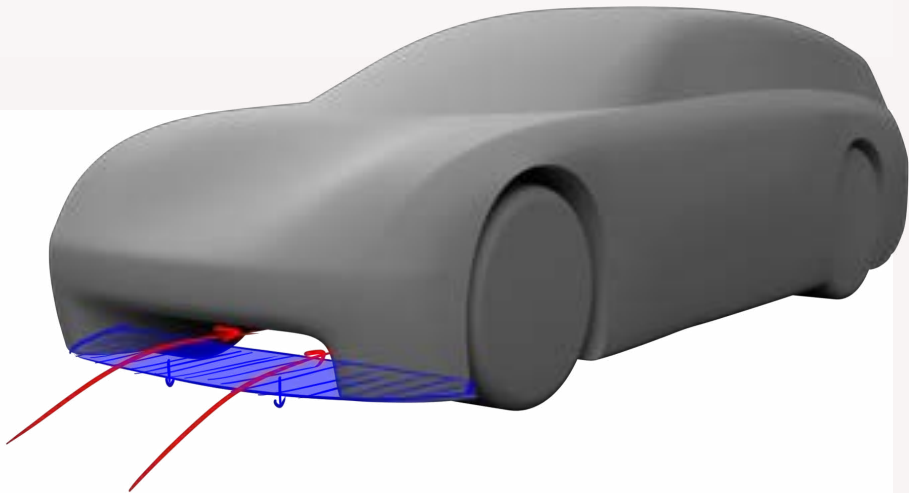
Overall, I think that the realistic structural layout of the vehicle means that it hit most of these requirements. Although, visibility may be the most compromised, especially at the rear of the vehicle. One way to tackle this is making the rear ‘cones’ an active element that could be stowed.

Concept Improvements

FRONT SPLITTER

In order to bring down the lift generated at the front of the car, a splitter could be added, pushing the nose of the car down.

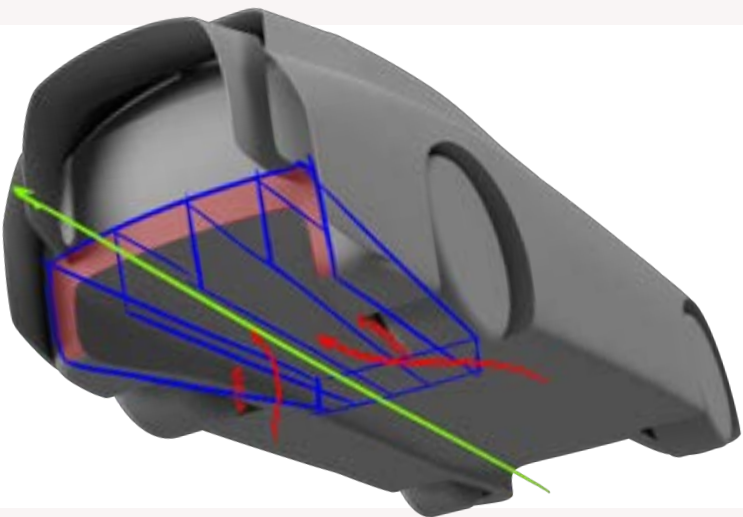
It would be important to monitor the effects of the splitter on the sloping intake for the Venturi tunnel, as it may disturb the inflow.



DIFFUSER SIZE

Increasing the size of the diffuser is another simple method to increase downforce.

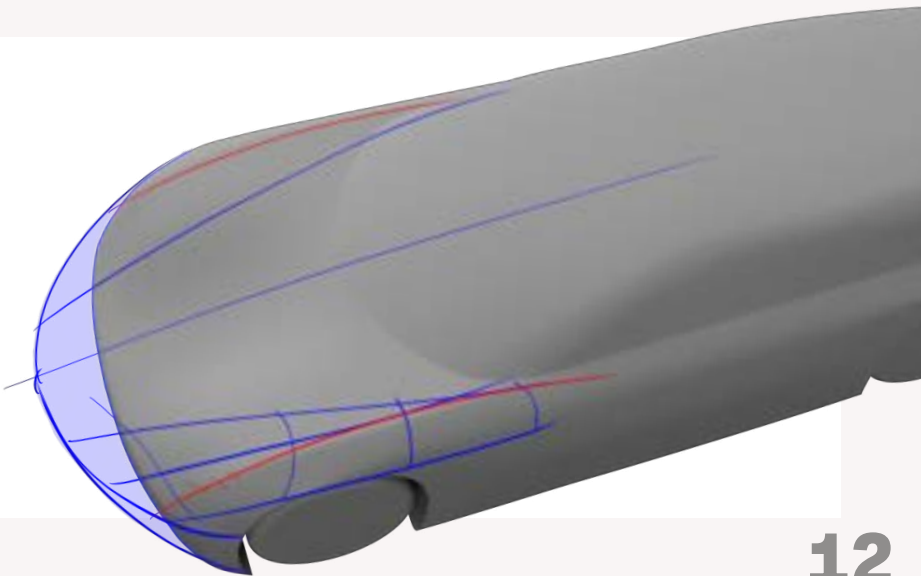
In addition to increasing the overall volume, I could add strakes to better manage the turbulent flow that can cross into the diffuser from the wheels and tires.

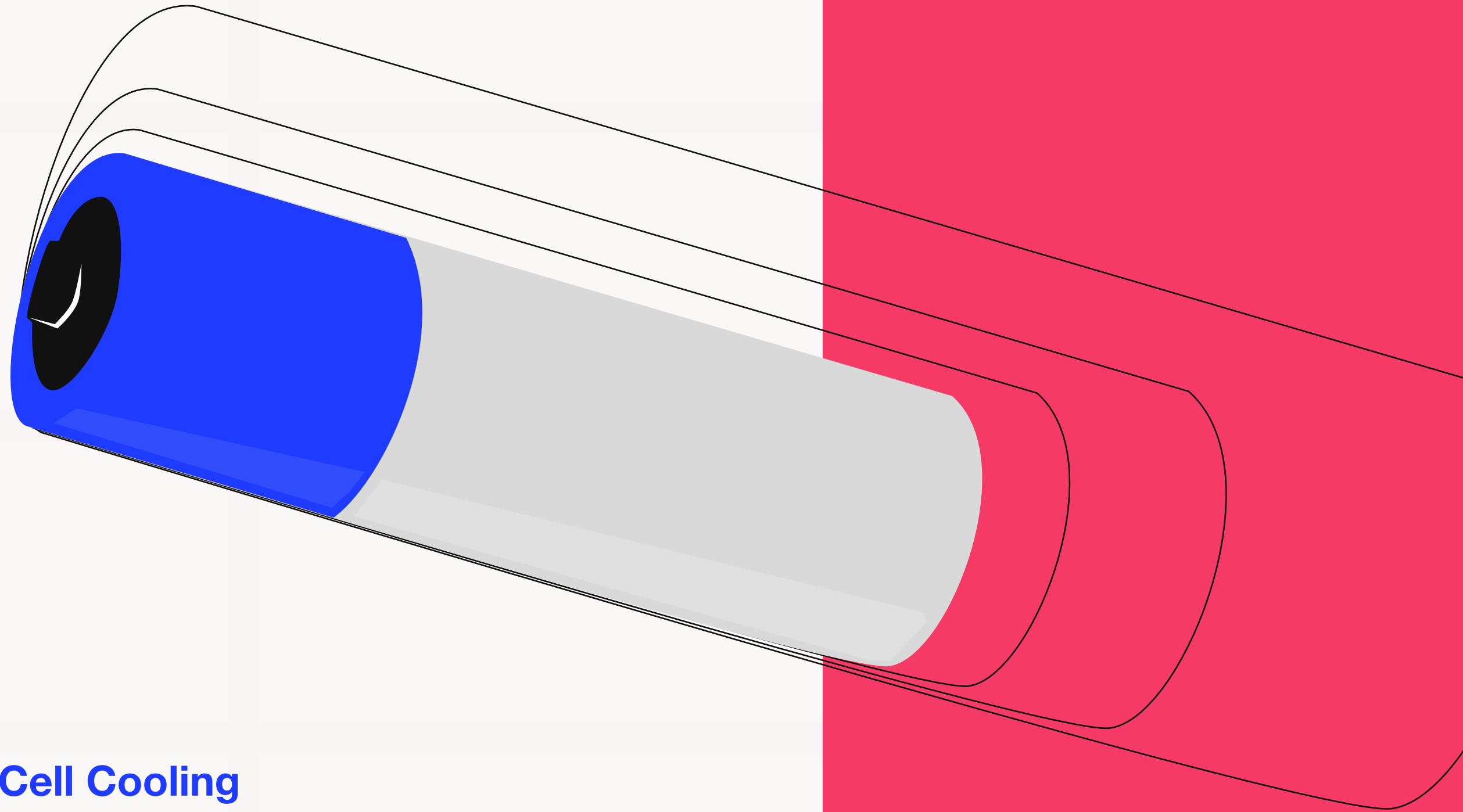


NOSE SHAPE

Further streamlining the nose shape could help to improve downforce and reduce drag, giving it a more curved and wing-like shape could help more air to flow over the vehicle.

As well as this, I would smooth the raised over-wheel sections to reduce any vorticity/turbulence that the sharp creases create.

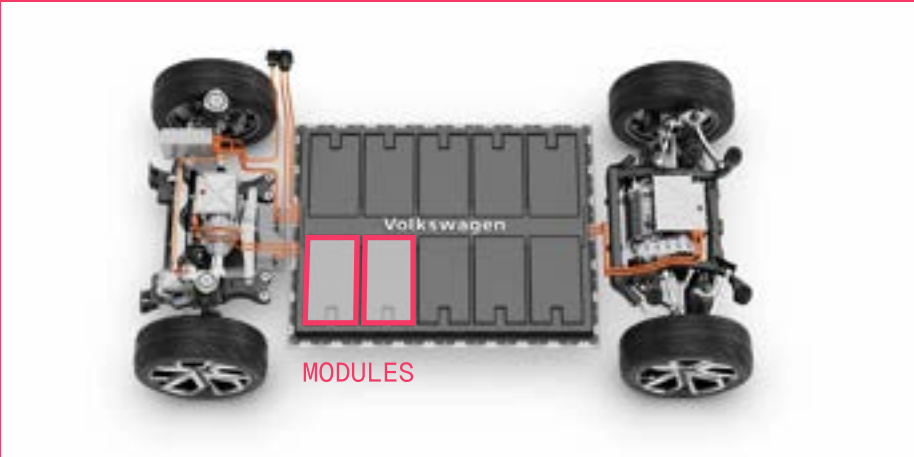




Pipe Fluidics and **Cell Cooling**

Research

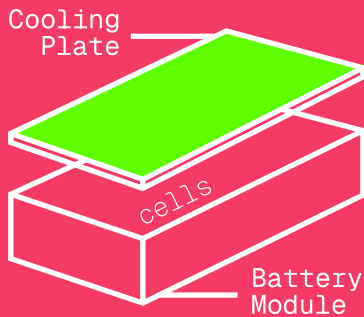
DE2 Thermofluids - Part 2



[11]

Construction + Layout

Battery Electric vehicles are most commonly constructed on a single battery pack. The layout is often a 'skateboard' platform like VW's MEB (above) which underpins 19 vehicles. These large battery packs are segmented into 'modules' of cells.



User Desires

FAST CHARGING

fast charging is essential for modern EV's, helping improve day to day practicality and reduce range anxiety. This is increasingly important as energy density and overall battery size grow.

LIFESPAN

The lifespan of an EV battery is extremely important, unlike many other electronic products, cars need to be able to last 10,15 or 20 years and possibly multiple users. Battery degradation can harm this, so it must be limited as much as possible

CELL PERFORMANCE

The performance of an EV depends heavily on the voltage the battery cells are able to produce. Cells must be kept between 20 and 30C for optimum output.

Current Issues

FAST CHARGING

[12]

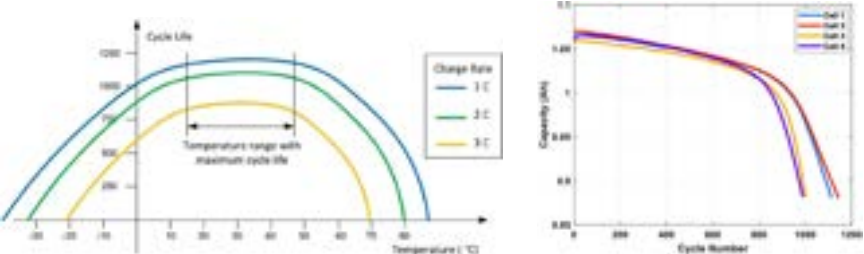


- Charging can increase temperatures to potential unsafe levels, often leading to 'thermal throttling' of charging speed and even damage to the battery.
- This can easily be caused by single point in battery due to on-uniformity in temperature regulation.
- Secondly, the heat transfer is important to dissipate heat as quickly as possible and keep charging speeds at sustained highs.

LIFESPAN

[13]

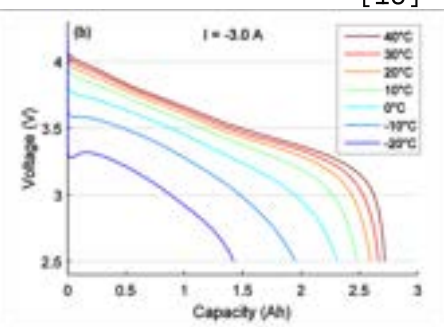
[14]



- Temperature dramatically effects cell cycle life - resulting in many less possible cycles when exposed to extreme high or low heat.
- This can likely be localised within a module/cell and will happen in certain modules before others.
- Cells also degrade rapidly once small imperfections have formed in the battery's structure.

CELL PERFORMANCE

[15]



- Cell performance is heavily effected by temperature.
- It effects both the capacity (Ah) of the cell as well as the voltage they are able to output.
- Changes in output voltages and capacities across a battery are difficult to manage
- This requires more computation from the BMS and is more prone to error, causing safety concerns.

Manufacturing Considerations



Cooling plates for EV battery packs are typically manufactured using stamped aluminium sheets that are then brazed for joining. This approach is lightweight and cost-effective for high-volume production, but limits the possible geometry and complexity.

- Minimum bend radius: Tight turns risk thinning and cracking during stamping.
- Channel depth limits: Restricted by forming capability and sheet stiffness.
- Pressure capability: Thin stamped constructions limit the max pressure of the system.
- Manifold complexity: Converging or multi-inlet/outlet regions increase sealing difficulty and distortion risk.

Design Objectives

PRIMARY OBJECTIVE



Reduce differences in temperature,improving uniformity across the module's surface.

SECONDARY OBJECTIVE



Increase heat transfer, increasing the rate that energy can be transferred from the battery's surface.

PRACTICAL CONSIDERATIONS

From the Module:

- Single inlet + outlet system
- 440*230*20 max cooling plate size

Other

- Simple cross sectional geometry for manufacturing
- Single-surface cooling plate design

Benchmarking

DE2 Thermofluids - Part 2

Ideal Cooling Plate

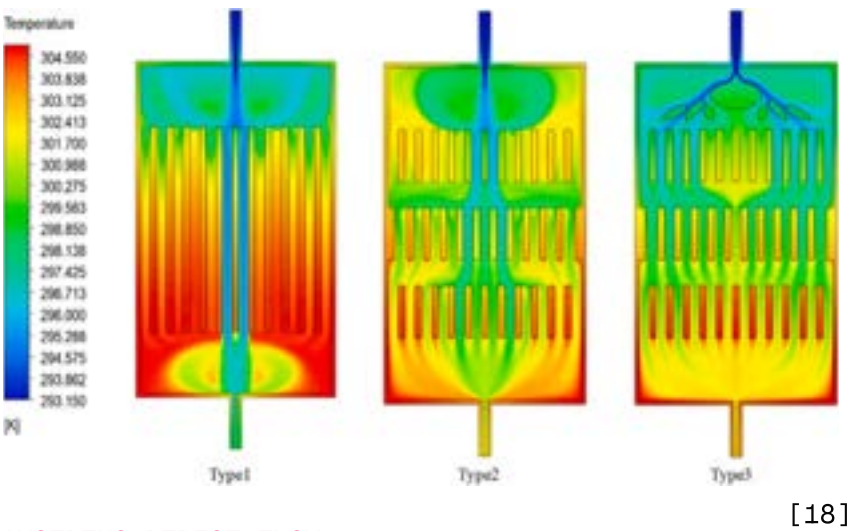
For ideal thermal performance, the cold plate layout should distribute coolant evenly from inlet to outlet. Near the inlet, flow should be guided to prevent bypassing and ensure early heat absorption. Through the centre, channels should maintain uniform coverage across the cells, and towards the outlet, the layout should avoid stagnation while sustaining consistent heat removal with minimal pressure loss.



Basic Principles

- Uniform flow distribution → Uniform temperature
- Higher velocity → Higher heat transfer (but ↑ pressure drop)
- Longer flow path → Higher coolant temperature rise
- More parallel paths → Lower pressure drop per path
- Pump power ∝ Flow rate × Pressure drop

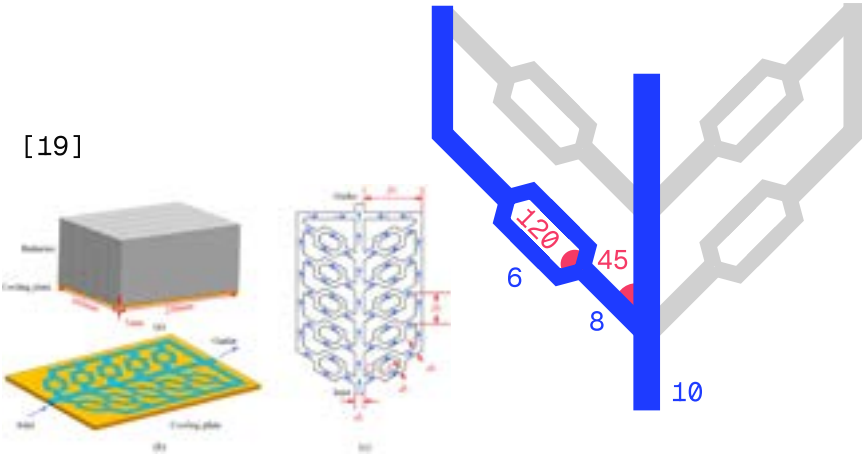
STAGGERED CHANNEL + FLOW VEIN



AVOIDING DIRECT FLOW

Using staggered channels is clearly very effective at increasing temperature uniformity. This layout forces the the coolant to flow more uniformly throughout the plate. The addition of the initial flow vein also dramatically increases the lateral spread, helping the flow reach the corners of the plate.

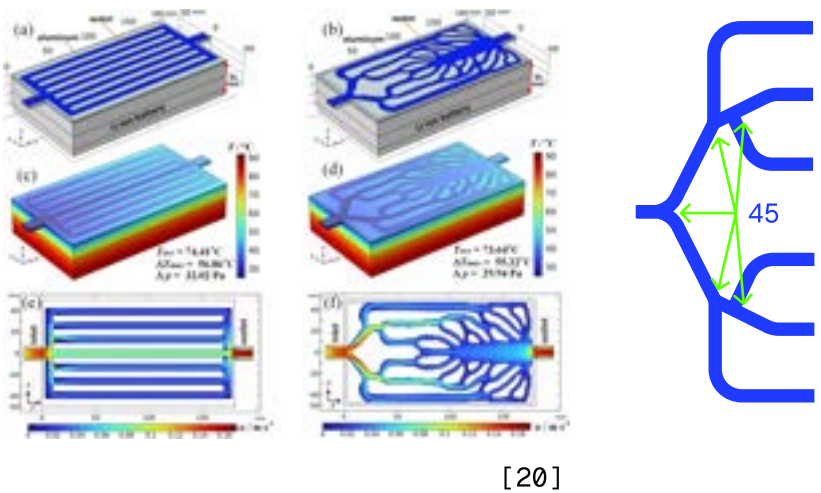
BIO-INSPIRED WING-VEIN



'POLYGON' SIDE CHANNELS

Inspired by vein structures in animal wings, these branches in side pipes aim to better distribute coolant over the surface without a large increase in pipe pressure. From this research it was found that the optimal angles were 45deg for the main branches, with 120deg separating the polygon sections.

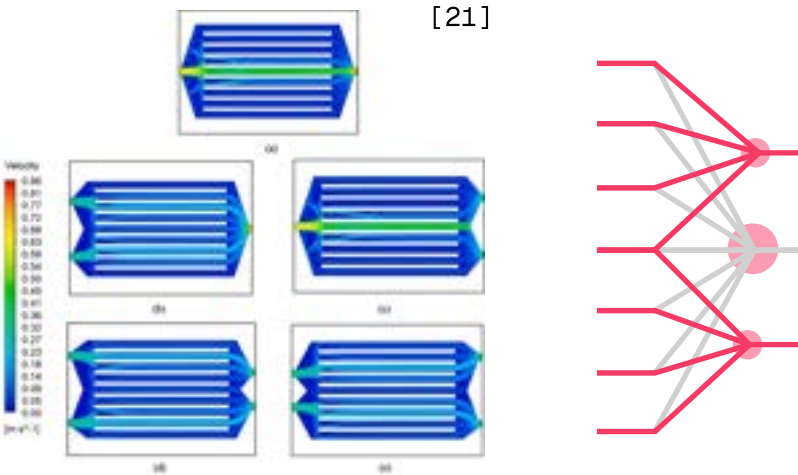
TOPOLOGY OPTIMISATION



INITIAL SPLITTING

This pipe layout was generated based on optimisations for temperature uniformity and pipe pressure dynamics. It successfully reduced maximum temps by over 1C while reducing pressure differences by over 2Pa. The most interesting part of the design is the initial splitting which only uses 45deg angle splits.

MULTI-INLET/OUTLET



DOUBLE OUTLET

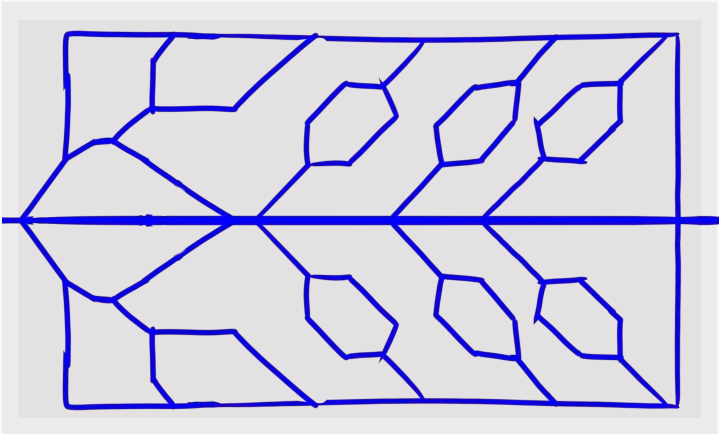
The main benefit of the double outlet configuration is reducing the stagnation and turbulence that combining many flows into 1 causes. By separating this out into 2 outlets, a more constant velocity is achieved, though this will increase pressure and both manufacturing + flow path complexity.

Ideation + Final Concept

DE2 Thermofluids - Part 2

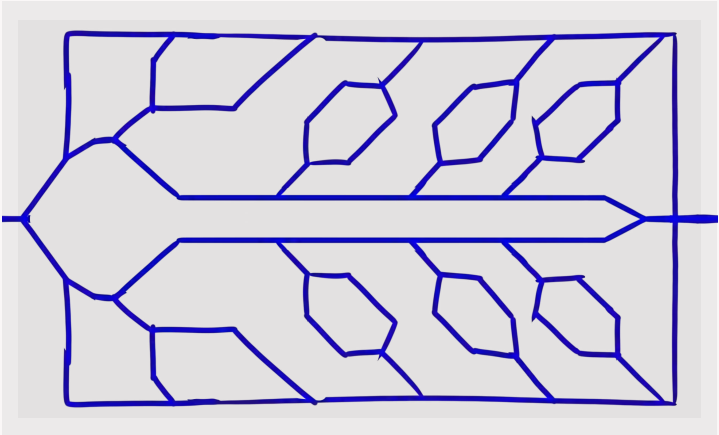
1

initial layout



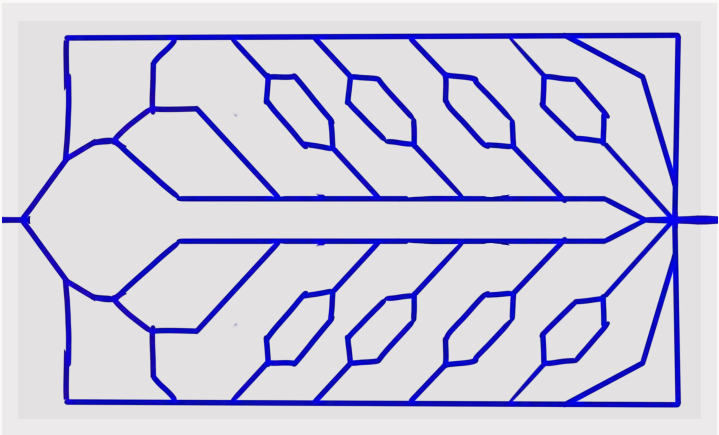
2

Removed Direct Flow

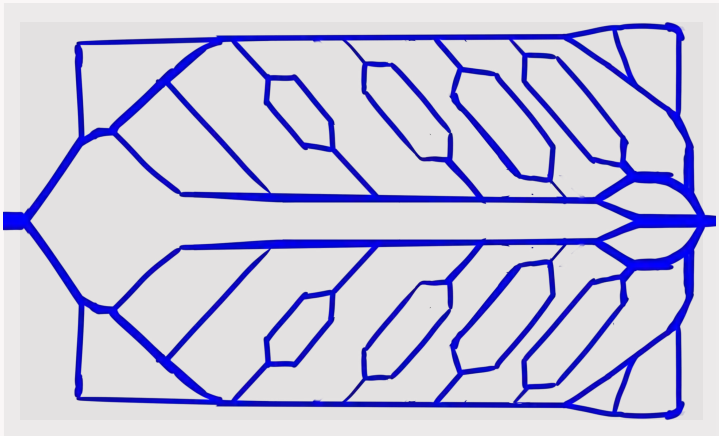


3

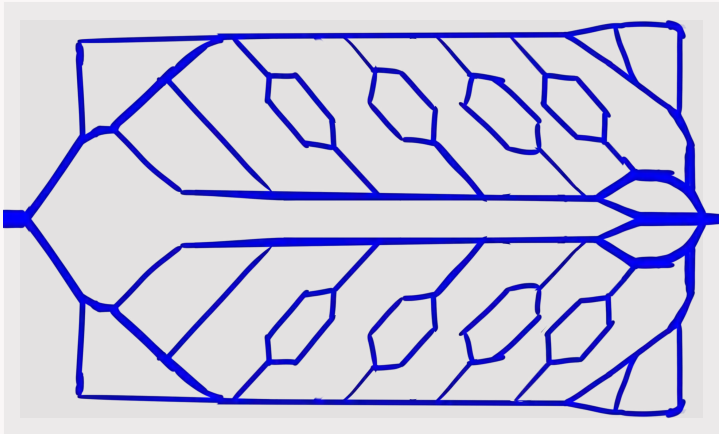
New Vein Direction



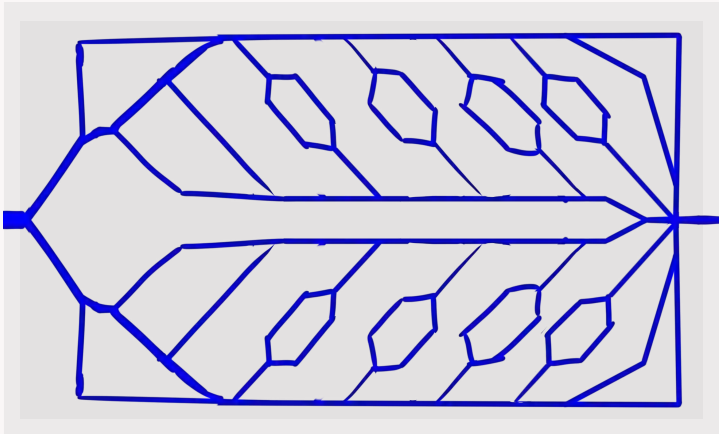
7 Scaling Vein Length



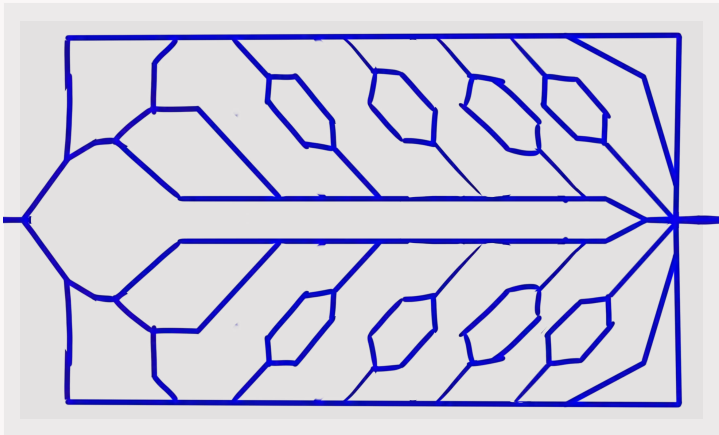
6 Split Outlet Join



5 Better Vein Distr.

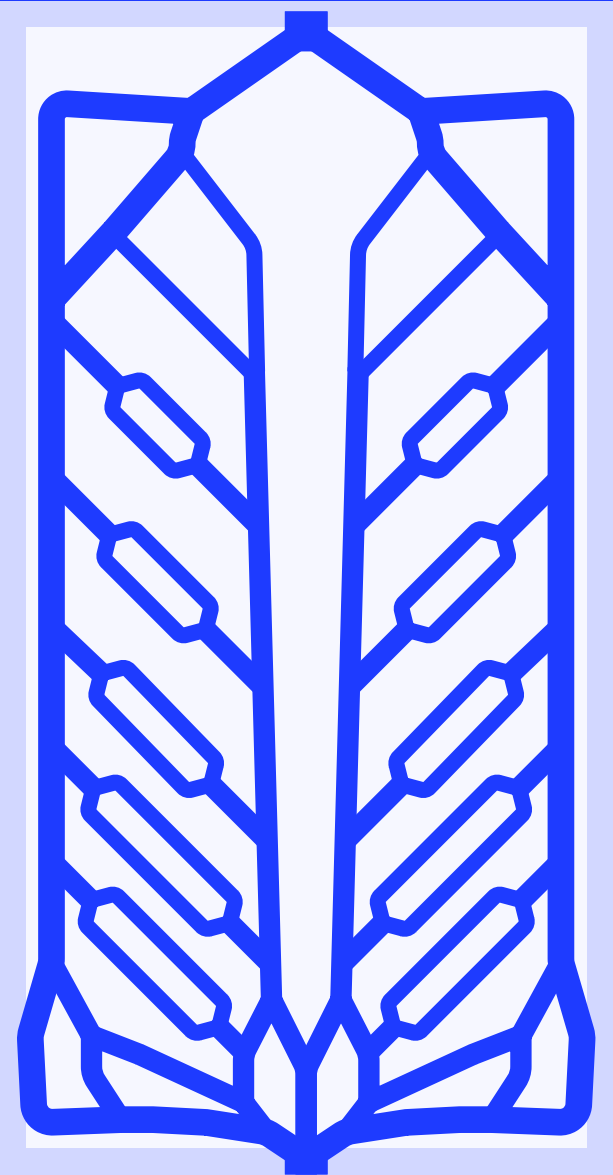


4 Better Vein Distr.



FINAL PIPE LAYOUT

INLET



OUTLET

- Pipe size limited to 3 specific diameters (10mm, 8mm, 6mm) and corner radii (3mm, 10mm) for manufacturing simplicity.
- Only 1 point of convergence with more than 2 pipes, to reduce complexity and flow stagnation.
- Little to no 'direct flow' or bypass flow-paths, increasing cooling coverage.

Appendix

DE2 Thermofluids - Part 2

REFERENCES - PART 1

[1] United States Environmental Protection Agency (EPA) (2021) Automotive Trends Report. Available at: <https://www.epa.gov/sites/default/files/2021-01/documents/420r21003.pdf> (Accessed: 19 February 2026).

[2] Society of Motor Manufacturers and Traders (SMMT) (2024) UK new car registrations data 2023. Derived statistic: Estate cars accounted for approximately 5-6% of UK new car registrations in 2023 (~110,000 of ~1.9 million registrations).

[3] ASME Digital Collection (n.d.) Figure reference. Available at: https://asmedigitalcollection.asme.org/view-large/figure/23638910/fe_139_02_021105_f009.png (Accessed: 19 February 2026).

[4] MDPI (2023) World Electric Vehicle Journal, 14(4), 236. Available at: <https://www.mdpi.com/2311-5521/5/4/236> (Accessed: 19 February 2026).

[5] TechRadar (2023) Mercedes-Benz Vision EQXX. Available at: <https://www.techradar.com/news/mercedes-benz-vision-eqxx> (Accessed: 19 February 2026).

[6] Porsche AG (2024) Porsche rounds off second Taycan generation. Available at: <https://newsroom.porsche.com/en/2024/products/porsche-rounds-off-second-taycan-generation-37832.html> (Accessed: 19 February 2026).

[7] Autoblog (n.d.) Used vehicle spotlight: Honda Insight. Available at: <https://www.autoblog.com/features/used-vehicle-spotlight-honda-insight> (Accessed: 19 February 2026).

[8] Cinch (2024) New Volkswagen ID.7 Tourer electric estate car. Available at: <https://www.cinch.co.uk/news/new-volkswagen-id-7-tourer-electric-estate-car> (Accessed: 19 February 2026).

[9] MDPI (2024) Energies, 17(17), 9693. Available at: <https://www.mdpi.com/2076-3417/15/17/9693> (Accessed: 19 February 2026).

[10] arXiv (2025) Paper 2504.01231. Available at: <https://arxiv.org/pdf/2504.01231> (Accessed: 19 February 2026).

REFERENCES - PART 2

[11] Sunrise Volkswagen (n.d.) MEB platform overview. Available at: <https://www.sunrisevolkswagen.com/meb-platform/> (Accessed: 19 February 2026).

[12] Raythink (n.d.) Infrared thermal cameras for EV battery safety. Available at: <https://www.raythink-tech.com/blogs/infrared-thermal-cameras-for-ev-battery-and-charger-safety/> (Accessed: 19 February 2026).

[13] ResearchGate (n.d.) Review and recent advances in battery health monitoring. Available at: https://www.researchgate.net/publication/260030309_Review_and_recent_advances_in_battery_health_monitoring_and_prognostics_technologies_for_electric_vehicle (Accessed: 19 February 2026).

[14] ResearchGate (n.d.) Capacity Prediction and Validation of Lithium-Ion Batteries. Available at: https://www.researchgate.net/publication/346714624_Capacity_Prediction_and_Validation_of_Lithium-Ion_Batteries-Based_on_Long_Short (Accessed: 19 February 2026).

[15] Green Tech Renewables (n.d.) How temperature affects battery performance. Available at: <https://www.greentechrenewables.com/article/how-does-temperature-affect-battery-performance> (Accessed: 19 February 2026).

[16] LinkedIn (2024) Chen, K. Design of BYD Seal CTB cold plate. Available at: <https://www.linkedin.com/pulse/design-byd-seal-ctb-cold-plate-keven-chen/> (Accessed: 19 February 2026).

[17] PVMARS (n.d.) Liquid cold plate types for Tesla Powerwall battery cooling. Available at: <https://www.pvmars.com/liquid-cold-plate-types-for-tesla-powerwall-battery-cooling/> (Accessed: 19 February 2026).

[18] ScienceDirect (n.d.) Article S129007292500804X. Available at: <https://www.sciencedirect.com/science/article/pii/S129007292500804X#fig11> (Accessed: 19 February 2026).

[19] ScienceDirect (n.d.) Article S0017931025015807. Available at: <https://www.sciencedirect.com/science/article/pii/S0017931025015807#fig0006> (Accessed: 19 February 2026).

[20] ScienceDirect (n.d.) Article S2352152X23017590. Available at: <https://www.sciencedirect.com/science/article/pii/S2352152X23017590#f0020> (Accessed: 19 February 2026).

[21] ScienceDirect (n.d.) Article S0142727X24003722. Available at: <https://www.sciencedirect.com/science/article/pii/S0142727X24003722> (Accessed: 19 February 2026).